

Agnes Namyalo

Data Scientist

✉ namyaloagie20@gmail.com ☎ +256778397172 📍 Kampala,Uganda Kampala

Education

Professional certificate in Data Science And Machine Learning

Mar 2023 - Jun 2023

Massachusetts Institute of Technology, Great Learning

Python for Data Science, Statistics for Data science, regression and prediction, Classification and Hypothesis testing, Deep learning, recommendation systems, Networking, and Graphical Models. The projects handled during this courses helped me comprehend important programming paradigms in Python and machine learning algorithms.

Msc in Mathematics of New Material Design,Nanotechnology

Aug 2019 - Oct 2020

Gdansk University of Technology, Gdansk, Poland

Computer modeling and design of materials, Mechanics of composites and metamaterials, Introduction to low dimensional systems and nanotechnology, Classical simulations with particles, Ethics in nanotechnology, Humanities and social science, Quantum simulations with particles.

Msc in Mathematical Engineering (scientific Computing)

Aug 2018 - Oct 2019

University of L'Aquila, Italy, L'Aquila,Italy

Control systems,Dynamical systems and bifurcation theory,Applied partial differential equations,Functional analysis in applied mathematics and engineering,Stochastic modeling and simulations. Discrete and continuum mechanics with applications,Complex analysis,Data analysis and data mining,Numerical methods for linear algebra and optimization

Bsc in Mathematics and Computer Science

Aug 2012 - Oct 2015

Busitema University, Busia,Uganda

Systems Analysis and Design,Probability and Statistics,Operating SystemsStatistical Data Analysis,Ordinary Differential Equations, Computer network and Data communications,Database Management Systems,Object Oriented Programming,Numerical analysis,Probability Theory,Computer Architecture,Simulation and Modeling,Stochastics Processes,Software Engineering,Data structures and Algorithms,Mechanics

Research Experience

- Worked with Dr Maciej Bobrowski at Gdansk University of Technology,Poland in the development of the Msc. Dissertation Research about the Parylene polymerization's termination reaction by burning in oxygen,quantum calculations.

Research Interests

- Scientific Computing, Applied Mathematics, Data science,Data analysis, Machine learning, Statistitcal analysis, Numerics and Applications, Mathematical modelling.

Technical Skills And Competences

- Python, R, SQL, Linux Shell, bussiness intelligence, Jupyter notebooks, Pandas, Tableau, Excel, Microsoft Power BI, Numpy, Mathematical knowledge, Data analysis, Statistical analysis, Sci-kit learn, Regression techniques, and Data visualization technique

- Communication

Languages

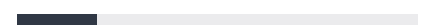
English



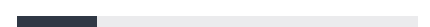
Luganda



Italian



Polish



Employment

Data Scientist

Jun 2022 – Present

Dfcu Bank, Kampala, Uganda

- I have used SSIS (SQL Server Integration Services) for data integration used SSIS tools and developed Python scripts to extract data from the core banking system and different sources to the Bank's data warehouse which increased data reliability hence resulting in more accurate forecasting and improved decision-making.
- As the operations team faced the challenge of carrying out reconciliation processes which took over 2 hours, I automated the Bank reconciliation process of Bank to wallet transactions to 3minutes, while building this automation, I used Power Automate to pick the emails automatically from Outlook and drop them to the designated folder then python script runs to extract, manipulate and reconcile the data from the files.
- DFCU Bank was facing a high churn rate which was impacting its revenue and profitability and needed to find ways to improve customer retention. I used Python and Python libraries for data science like pandas, NumPy, and matplotlib to perform an exploratory analysis of the customer data to understand their behavior and demographics. I then used machine learning techniques to develop a predictive model using a logistic regression and Random Forest that could identify customers at risk of churning. Based on this analysis, I identified several key factors that were contributing to churn such as low engagement. As a result of these interventions, the company was able to reduce its churn rate and increase customer lifetime value.

Business Intelligence Analyst

Jan 2022 – May 2022

Equity Bank, Kampala, Uganda

- Integrated BI assets into customer relationship management tasks, improving service personnel's available intelligence.
- Collected data from wide-ranging sources.
- Maintained proficiency with latest database technologies.

Data Scientist

Jun 2021 – Oct 2021

Centre For Computational Biology–Uganda Christian University, Kampala, Uganda

- Creating models about future perspective of COVID-19 pandemic in Uganda.

Project: Extracted data for Uganda COVID-19 cases from online website(Our World in Data). Created visuals using python libraries like pandas, matplotlib to give statistical analysis on comparison with changes in weekly,monthly and regional number of COVID-19 cases. Created a time series forecast model of future cases using prophet library, regression model to make predictions of COVID-19 cases to help in making informed decisions on what to expect and plan regarding the pandemic state in Uganda.

Mathematics and Computing Instructor

Jan 2016 – Aug 2018

Mathematics and Computing Instructor, Bombo, Uganda